PE Rubric 1: The Nervous System Exam Rubric with Special Tests 26-27

Communication, Professionalism, and Patient Care
Introduced self, asked the patient name, preferred way to be addressed, DOB, and Age
Sanitized hands with air dry or washed hands at sink for 20 seconds, before **AND** after touching the patient and when appropriate
Obtained consent for the patient exam, acknowledged the patient's ability to relinquish consent at any time, and obtained permission to move draping to touch or examine skin (especially in sensitive areas) when needed throughout the exam
Supported the patient with comfort and empathy measures during exam: Checked to make sure that the exam was comfortable, elevated or lowered the exam table, offered assistance with changes in position, and used proper draping (if any of these things apply to the testing situation)
Spoke only to the patient during the exam, explained what was necessary, and used nonmedical jargon to explain things
Was respectful and professional in manner and demeanor throughout the exam to the patient and the faculty facilitator proctor
Parts 1 and 3= General Neurologic Screening Exam (Mental Status-Motor with Gait/Stance/Coordination/Strength-Sensory-Reflexes-and Cranial Nerves)
Part 1- Mental Status, Gait/Stance, Coordination, Sensory, Motor, and Reflexes
Part 1- General Appearance and Mental Status of the Patient: This includes many things to consider: **Appearance and Behavior** (Dress, Motor Behavior/Posture, Level of Consciousness, Facial Expression, Manner, Affect, and Relationships to things), **Speech and Language** (Fluency/Quantity/Rate/Volume, and Articulation), **Mood** (Sad/Happy/Euphoric/Anger/Depression/Anxiety), **Cognition** (Recall, Orientation to Person/Place/Time/and Situation, Attention, Memory being remote and recent), **Thought Process and Content** (Logical, Organized, Relevant, Coherent vs Obsessions, Compulsions, Anxieties, Phobias, or Delusions); Secondary Perception, Insight, and Judgement
Gait Analysis: Ask the patient to walk normally (shoes off) **AND include one other**: Rise from chair without using arms, **OR** Tandem/heel to toe walking *(If the patient is in a hospital bed or has different mobility, document bedrest, observed mobility & device.)*; **Optional** Toe walking (on tip toes) **AND** Heel walking (on heels)
Coordination (Cerebellar Screening): Ask patient to perform 1 for UE AND 1 for LE, finger to nose **OR** rapid alternating hand movements **BL AND** heel down shin **OR** ball of foot tapping **BL**
Sensory Screen: Explain to the patient with eyes open, but test with eyes closed; Use **Light Touch** (a cotton wisp or a light static finger touch) to test the 4 extremities distal vs proximal, 2 touches per limb (8 touches) **BL** and have the patient point to the touch when felt (also, do not touch 2 places at the same time BL)
UE Motor- Myotomes (nerve root strength of certain muscle groups against resistance which are stated as intact/full strength or not intact/any weakness; muscle strength otherwise would need to be assessed on a 0-5/5 grading scale): C5 shoulder abduction, C6 elbow flexion **OR** wrist extension, C7 elbow extension **OR** wrist flexion, C8 grip **OR** finger flexion **AND** T1 finger abduction **AND** adduction **OR** C8-T1 finger and thumb opposition **BL**
LE Motor- Myotomes (nerve root strength of certain muscle groups against resistance which are stated as intact/full strength or not intact/any weakness; muscle strength otherwise would need to be assessed on a 0-5/5 grading scale): L 2 hip flexion, L3 knee extension, L4 foot dorsiflexion, L5 big toe extension **AND** S1 foot eversion **OR** foot plantarflexion **BL**
UE Deep Tendon Reflexes (stated as intact if 2/4 or DTRs are graded 0-4/4 if not 2/4): C5 Biceps, C6 Brachioradialis **AND** C7 Triceps reflexes **BL**
LE Deep Tendon Reflexes (stated as intact if 2/4 or DTRs are graded 0-4/4 if not 2/4): L4 Patellar Tendon **AND** S1 Achilles Tendon reflexes **BL**
Plantar Response: L5, S1 checks for Babinski testing by using the metal end of the reflex hammer and stroking the foot from the heel up and across (7) **BL**
Abdominal Reflexes (Cutaneous or Superficial Stimulation): T8, T9, T10 and T10, T11, and T12 (not always performed)
Part 2- Neurologic Case Specific Special Testing (Spinothalamic Tracts, Dorsal ML/Posterior Columns, and Cortex):
Test Spinothalamic Tracts for **Pain- Sharp** (use a toothpick, end of a paperclip, or end of a wooden stick of a cotton tip swab) **vs Dull Touch** (use a cotton wisp, a light touch with a cotton tip swab, or a light static finger touch, without using a swipe approach) **OR Temperature- Hot vs Cold BL:** Explain to the patient with eyes open, but test with eyes closed and ask the patient to explain the touch or temperature and point **BL.** Test the 4 extremities distal vs proximal, 2 touches per limb (8 touches) **BL** (also, do not touch 2 places at the same time BL)
Test Dorsal Column ML for **Position OR Vibration Sense OR Light Touch:** Explain to the patient with eyes open, but test with eyes closed and move the patient's fingers, toes, or joints up or down and ask the position **OR** place a vibrating tuning fork on the joint and ask if vibration is felt and when it stops **OR** use a cotton wisp or a light static finger touch to test the 4 extremities distal vs proximal, 2 touches per limb (8 touches) **BL** and have the patient point to the touch when felt (also, do not touch 2 places at the same time BL)

Test for the Dorsal Columns (proprioception/position with help from other systems like visual and vestibular) for Static Screening: Romberg Test: Ask the patient to stand with feet together and close eyes to see if the patient sways in 30-60 seconds (make sure to protect patient); Note that if the patient can't do this with eyes open, stop the test before performing with the patient eyes closed
Test for Position, Stance, Static Screening, Spasticity, and Corticospinal Tract issues: Pronator Drift Test: Ask the patient to hold arms out horizontally, palms up, with eyes closed to see if palms or arms drift inward and down; Briskly tap arms downward to see if the arms drift or have writhing movements away
Test for Both Tracts and Cortex: Two Point Discrimination Test: Explain to the patient eyes open, but test with eyes closed for 2 point discrimination **BL**
Test for Both Tracts and Cortex: Stereognosis Test: Put a common object in each of the patient's hands with eyes closed and have the patient identify the objects
Test for Both Tracts and Cortex: Graphesthesia Test: Draw a number on each of the patient's hands with eyes closed and have the patient identify the numbers **BL**
Dermatomes UE (sensory root of a single spinal nerve): Use **Light Touch** (a cotton wisp or a light static finger touch) to test C5 lateral upper arm, C6 lateral forearm/wrist and dorsum of thumb, C7 dorsum of middle finger, middle palm and middle fingers, C8 dorsum of little finger to medial hand and wrist, T1 medial forearm **BL (ASIA sensory points are underlined)**
Dermatomes LE (sensory root of a single spinal nerve): Use **Light Touch** (a cotton wisp or a light static finger touch) to test L1 anterior inguinal area and posterior upper lower back, L2 anterior upper middle thigh and posterior upper lower back, L3 medial knee to medial thigh and posterior back, L4 medial malleolus to knee to lateral leg and posterior lower back, L5 interspace between toes 1 and 2 to lateral lower leg and posterior lower back, S1 lateral malleolus to posterior leg and posterior lower back and upper sacral area **BL (ASIA sensory points are underlined)**
Upper Motor Neuron Lesion or Cord Compression Tests: Hoffman's/Tromner's Sign: Have the patient put the index finger and thumb of the same hand a couple of inches apart, flick the **3rd/Middle finger (down with Hoffman and up with Tromner)** and see if the if the index finger & thumb move closer together, which would be a positive test **OR** perform a **Babinski's Sign** (Plantar Reflex Response): A positive test is when the plantar surface of the foot is stroked with a noxious stimulus/probe as if writing a 7 and the toes go up which is positive vs. down which is a negative test **BL**
Meningeal Irritation Signs: Brudzinski's Sign: Patient supine, flex the patient's head to see if there is a pain response with the patient involuntarily flexing the hips & knees **OR Kernig's Sign:** Patient supine, flex the patient's hips & knees to 90 degrees to see if there is pain or resistance c extension of the knee beyond 135 degree
Cord Compression or Multiple Sclerosis Pain Radiation: Lhermitte's Sign: Flexing the head for the chin to touch the chest, causes pain down the spine and possibly the extremities is positive
Part 3- Cranial Nerve Exams: I-XII
CN 1 Olfactory, Smell, Sensory: Explain to the patient that you will be checking the sense of smell with the patient's eyes closed; Have the patient occlude one nostril at a time and ask the patient to identify the scent that is placed in front of each nostril **BL** (Note: change the scent for each nostril)
CN 2 Optic, Vision, Sensory: Peripheral Field Testing Test each eye of the patient, one at a time, in comparison of physician's peripheral vision **BL** with one eye closed or hand over one eye. Must achieve 4 quadrants with finger movement **OR** numbers (using the numbers 1, 2, and 5); Student must be at eye level with patient and about arm's length 2-3 feet apart from patient each eye & one eye covered at a time **BL**
CN 2 Optic, Vision, Sensory: Use the **Snellen OR Rosenbaum Chart** (at the appropriate chart distance) and with one eye covered at a time, ask the patient to read the smallest line that can be read **BL AND** then with both eyes open together (Note with or without glasses or contact lenses in place)
CN 2 Pupillary Reflex, Sensory and CN 3 Motor: Note the size and shape of the patient's pupils, then **Shine Direct Light** into each eye for pupil reflex **BL AND** check **Consensual Light** for each eye's pupil reflex **BL AND** check **Far to Near pupils** by asking the patient to follow your finger in from a distance of 16 inches far to near with both eyes to the nose and note: constriction/convergence/accommodation
CN 2 Evaluates eye red reflex with ophthalmoscope and performs Fundoscopic Exam: (must be able to turn on, hold the ophthalmoscope with finger on diopter wheel for focusing, and get light into each pupil) **BL** (look for red reflex, retinal vessels, cup and disc)
CN 3, Oculomotor (pupil constriction, lid elevation, and moves the eyes medial/inward and most other movements), **CN 4 Trochlear** (moves the eyes down and in), **AND CN 6 Abducens** (moves the eyes lateral/outward), **Extra Ocular Eye Movement LR6 (SO4)3:** The patient is instructed to follow the physician's finger with both eyes in at least the 6 cardinal gaze positions **with an H**; Look for symmetry, weakness, ptosis, and nystagmus vs. fluidity
CN 5 Trigeminal, Sensory V1 ophthalmic, V2 maxillary, V3 mandibular: Explain with the patient eyes open first, but test with the patient eyes closed in all divisions **BL** with sharp vs dull; **AND Motor:** Temporal and Masseter palpation of movement with jaw clenching teeth **OR** Pterygoid with lateral jaw movement vs resistance **BL**; **CN 5 and CN 7 Corneal Reflex Sensory and Motor for irritation and blinking with cotton wisp touching the cornea will not be tested**

CN 7 Facial, Motor: Ask the patient to Raise Eyebrows, Close Eyes Tight, Frown, Smile, Show Upper and Lower Teeth, **AND** Puff Cheeks **BL**; **Sensory: Taste salty, sweet, sour, and bitter on the anterior 2/3 of the tongue will not be tested**
CN 8 Vestibulocochlear, Hearing, Sensory: Explain with the patient eyes open, but perform with patient eyes closed, and stand behind patient unless mouth is covered (because the patient can feel the movement or read the lips): **Finger Rub** by one ear and have the patient point to which ear (do a couple of times) **OR Whisper a Word** and have the patient repeat the word (use a different word for each ear) **BL**
CN 9 Glossopharyngeal, Palate Elevation, Motor: Ask the patient say "Ah" or yawn (and look in the mouth with a light) **OR** Ask the patient to swallow and palpate the neck while the patient swallows; Sometimes the **GAG reflex is checked for coma patients etc. but will not be tested**; **Sensory: Taste will not be tested**
CN 10 Vagus, Palate Elevation, Motor: Speech is not hoarse with having the patient talk; Ask the patient to say " Ah" or yawn (and look in the mouth with a light) **OR** Ask the patient to swallow and palpate the neck; **Sensory: Pharynx and Larynx; The Gag Reflex will not be tested**
CN 11 Spinal Accessary, Motor: Trapezius: Shoulder Elevation vs. resistance **OR** Sternocleidomastoid: Head Rotation vs. resistance **BL** (should be 5/5 **BL**)
CN 12 Hypoglossal, Motor: Tongue movement, protrusion- move right and left **OR** push tongue against cheeks **AND** articulation of speech (have the patient speak)
Part 4- Neurologic Case Specific Special Testing for Cranial Nerves: Choose at least 3 or More Best Special Tests to Perform based on Symptoms
CN 1 Changes in the Sense of Smell: Is in the cranial nerve rubric above, but it is not usually tested unless the patient has lost the sense of smell **BL**
CN 2 Changes in Vision: Peripheral Field Testing is in the cranial nerve rubric above, but it is not usually tested unless the patient has vision loss or had a stroke
CN 2 Changes in Vision: Fundoscopic Exam with Red Reflex in each eye (must hold the ophthalmoscope properly with finger on diopter wheel and see light in eye)
CN 8 Changes in Hearing: Must perform both Weber (the vibrating tuning fork is placed on the top of the head for lateralization of vibration or sound) **AND Rinne Testing** (the vibrating tuning fork is placed on the mastoid bone and then in front of ear and ask the patient which sound is louder 1 or 2; Bone vs. Air); check **BL**
Vertigo Testing: Benign Paroxysmal Positional Vertigo (BPPV): **Dix-Hallpike Maneuver:** The patient sits upright with the head turned 45 degrees horizontally to the affected side and eyes wide open (you should see a fast torsional beat to that side); Then the patient lies back on the table in a supine position with the head off the table yet supported and turned with the ear pointed downward to the affected side and stay there for 1-2 minutes and the eyes are checked for nystagmus or a fast beat to that side **OR** the **Supine Head Roll Maneuver:** The patient is supine with the head elevated to 30 degrees and rotate the head to the right and then to the left; A positive finding is a horizontal nystagmus with the fast beat to the affected side
Headache, Sinus Pressure, AND/OR Congestion: Palpates **OR** Percusses **OR** Trans illuminates **BOTH** the Frontal **AND** Maxillary Sinuses **BL**
Headache, Check the Blood Pressure: The patient is relaxed, sitting upright with back supported, feet flat on the floor (uncrossed), with appropriate cuff size used and arm horizontal at heart (chest) level and checked **BL**